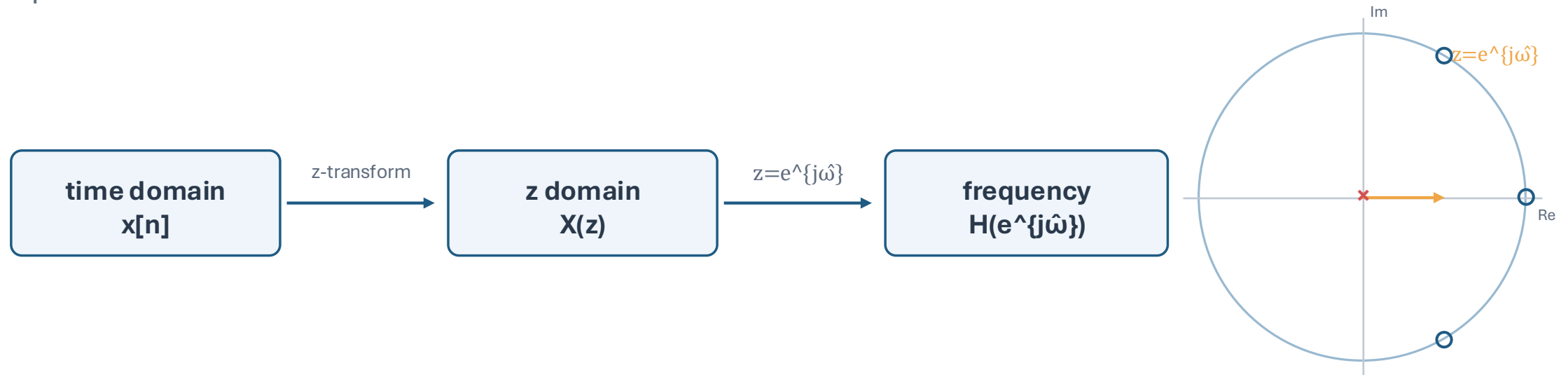


Topic 7

The z-Transform

From sequences and delays to polynomials, poles, zeros, and frequency response



Key idea. The z-transform turns time-domain filtering into algebra: delays become powers of z^{-1} and convolution becomes multiplication.

Definition of the z-transform

A sequence becomes a polynomial in z^{-1}

$$x[n] = \sum_k x[k] \delta[n - k]$$

$$X(z) = \sum_n x[n] z^{-n}$$

finite sequence $\Leftrightarrow$ finite polynomial in z^{-1}



$$X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \dots$$

- Each sample multiplies the matching power of z^{-1} .
- Missing powers simply have coefficient 0.

Key idea. For finite sequences, the z-transform is just a polynomial whose coefficients are the samples.

Example 1: sequence $\rightleftarrows$ z-transform

Read the coefficients of powers of z^{-1}

$$x[n] = \{ 3, 1, 4, 0, 5 \} \text{ starting at } n=0$$

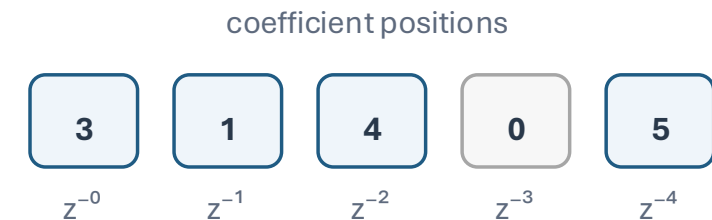
1 Write one term per sample: $3, 1z^{-1}, 4z^{-2}, 0z^{-3}, 5z^{-4}$.

2 Combine the terms into a polynomial in z^{-1} .

$$X(z) = 3 + z^{-1} + 4z^{-2} + 5z^{-4}$$

3 The missing z^{-3} term means $x[3]=0$.

4 A shifted impulse $\delta[n-5]$ has z-transform z^{-5} .



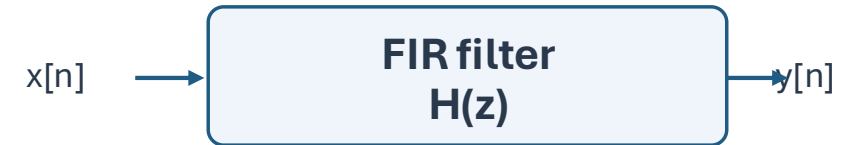
The z-transform of an FIR filter

The system function is the z-transform of $h[n]$

$$y[n] = b[0]x[n] + b[1]x[n-1] + \cdots + b[M]x[n-M]$$

$$h[n] = \sum_{k=0}^M b[k] \delta[n - k]$$

$$H(z) = \sum_{k=0}^M b[k] z^{-k}$$



Equivalent descriptions



Key idea. For an FIR filter, the coefficient list, impulse response, difference equation, and $H(z)$ contain the same information.

Time shifting and the unit-delay operator

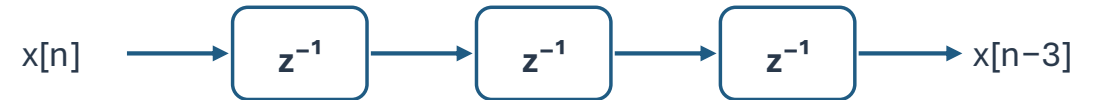
A delay becomes multiplication by z^{-1}

$$x[n-n_0] \Leftrightarrow z^{-n_0} X(z)$$

$$\delta[n-1] \Leftrightarrow z^{-1}$$

Interpretation:

- one z^{-1} block = one-sample delay
- m cascaded blocks = m-sample delay
- useful in block diagrams and difference equations



Key idea. The z-transform makes delay algebraic, so block diagrams can be read as polynomials.

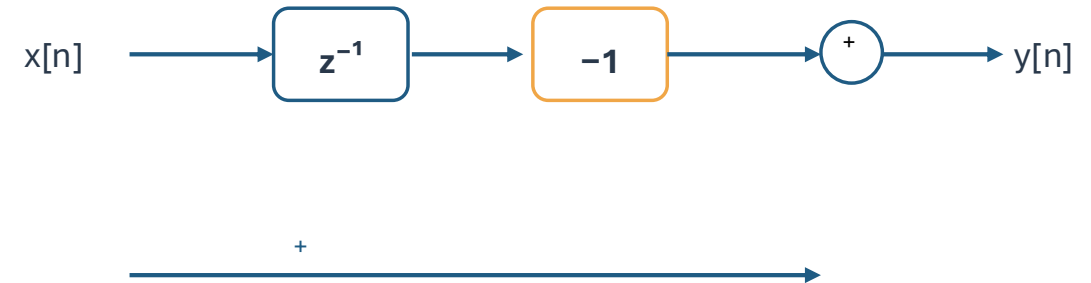
Operator notation in block diagrams

z^{-1} is a delay operator

$$y[n] = x[n] - x[n-1]$$

$$Y(z) = (1 - z^{-1})X(z)$$

$$H(z) = 1 - z^{-1}$$



This is the first-difference filter: present sample minus delayed sample.

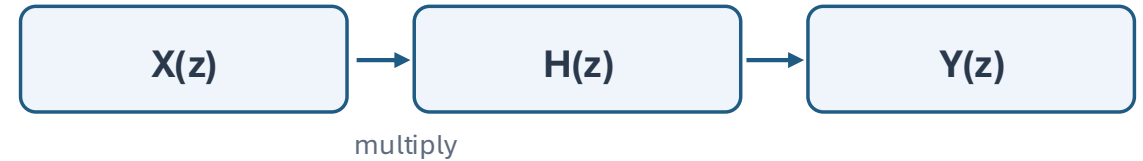
Key idea. Replacing a unit delay by z^{-1} turns block diagrams into algebraic expressions.

Convolution and the z-transform

Time-domain convolution becomes polynomial multiplication

$$y[n] = x[n] * h[n]$$

$$Y(z) = X(z) H(z)$$



Why this matters:

- Convolution output samples are coefficients of a polynomial product.
- Cascaded filters multiply their system functions.
- Factorising $H(z)$ decomposes a high-order filter into smaller sections.

$$(1+2z^{-1}+z^{-2})(1-z^{-1}) = 1+z^{-1}-z^{-2}-z^{-3}$$

Key idea. The z-transform turns the convolution operation into ordinary algebra.

Example 2: polynomial convolution

Multiply in the z-domain and read the coefficients

$$x[n]=\{1,2,1\}, \quad h[n]=\{1,-1\}$$

1 Write the z-transforms: $X(z)=1+2z^{-1}+z^{-2}$, $H(z)=1-z^{-1}$.

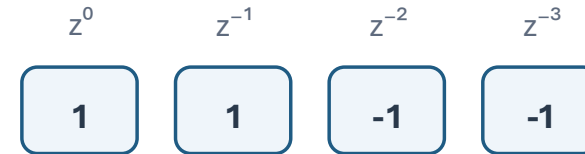
2 Multiply the polynomials in z^{-1} .

$$Y(z) = (1+2z^{-1}+z^{-2})(1-z^{-1})$$

$$Y(z) = 1 + z^{-1} - z^{-2} - z^{-3}$$

3 Read the output samples from the coefficients.

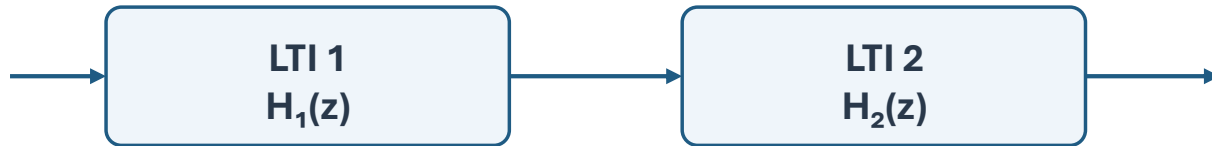
$$y[n] = \{1, 1, -1, -1\}$$



Key idea. Polynomial multiplication and finite convolution produce the same coefficient sequence.

Cascaded systems and factorisation

Products of smaller filters make larger systems



$$H_{eq}(z) = H_1(z)H_2(z)$$

$$h_{eq}[n] = h_1[n] * h_2[n]$$

Factorisation goes the other way:

$$H(z) = (1 - z^{-1})(1 + 2z^{-1})$$

A higher-order polynomial can be implemented as a cascade of lower-order sections.

Key idea. Cascades multiply in the z-domain; factorisation turns algebra into implementation structure.

Example 2: cascade and equivalent FIR

Multiply system functions, then read coefficients

$$H_1(z)=1-z^{-1}, \quad H_2(z)=1+2z^{-1}$$

- 1 Overall response is the product of the two sections.

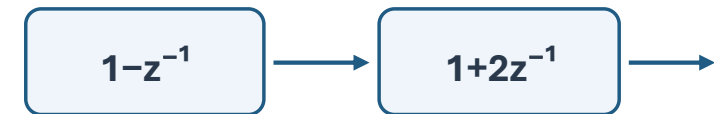
$$H_{eq}(z) = (1-z^{-1})(1+2z^{-1})$$

- 2 Expand the polynomial.

$$H_{eq}(z) = 1 + z^{-1} - 2z^{-2}$$

- 3 Read the equivalent impulse response / coefficients.

$$h_{eq}[n] = \{1, 1, -2\}$$



Key idea. The same result is obtained if the sections are reversed because polynomial multiplication is commutative.

Inverse filtering and deconvolution

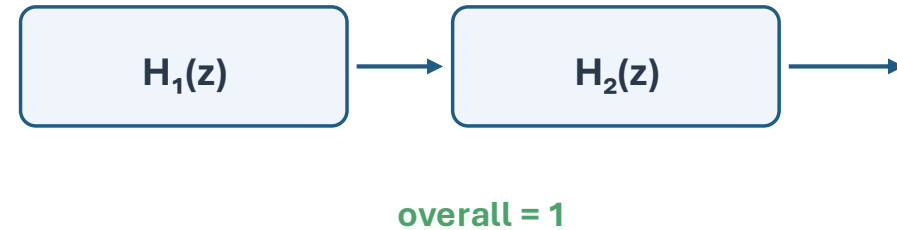
Undoing a filter means multiplying by its reciprocal

$$H_1(z)H_2(z) = 1$$

$$H_2(z) = 1 / H_1(z)$$

Important consequence:

- The inverse of a finite FIR polynomial is often a rational function.
- Its power-series expansion may be infinite.
- An exact causal inverse is often IIR, not FIR.



$$1/(1-az^{-1}) = 1 + az^{-1} + a^2z^{-2} + \dots$$

Key idea. Deconvolution is algebraically simple, but exact inverses of FIR filters are often infinite-length systems.

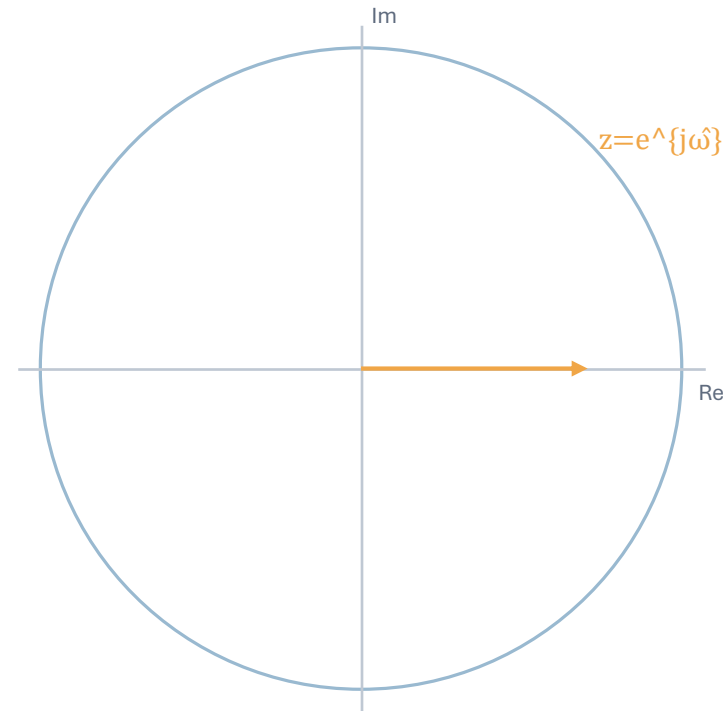
Relation between z-domain and frequency domain

Evaluate $H(z)$ on the unit circle

$$z = e^{j\hat{\omega}}$$

$$H(e^{j\hat{\omega}}) = H(z) \big|_{z=e^{j\hat{\omega}}}$$

- $\hat{\omega}$ is normalized radian frequency.
- As $\hat{\omega}$ moves from $-\pi$ to π , z moves once around the unit circle.
- Zeros on the unit circle cause frequency-response nulls.



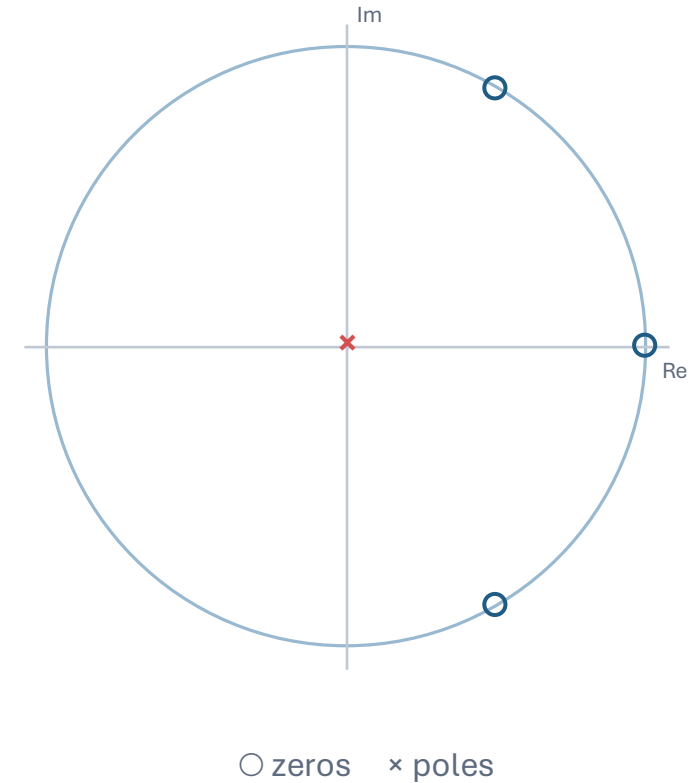
Key idea. The frequency response is the trace of the z-domain system function around the unit circle.

Poles and zeros of $H(z)$

Polynomial factors reveal where the response vanishes or blows up

$$H(z) = K \prod_i (1 - z_i z^{-1}) / \prod_r (1 - p_r z^{-1})$$

- Zeros z_i make $H(z_i)=0$.
- Poles p_r make $H(z)$ undefined.
- Finite FIR filters have zeros plus poles at the origin from negative powers of z .
- Zeros on the unit circle create exact frequency nulls.



Key idea. Pole-zero diagrams make the frequency-selective behaviour of a filter visible.

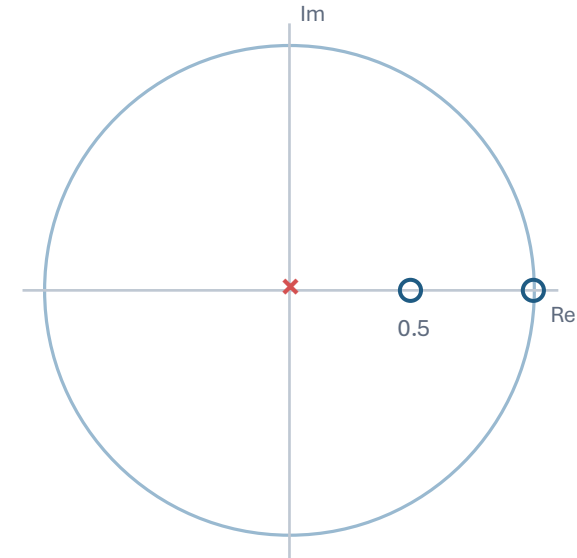
Example 3: from $H(z)$ to poles and zeros

Factor the polynomial and place the roots

$$H(z) = 1 - 1.5z^{-1} + 0.5z^{-2}$$

- 1 Multiply by z^2 to obtain an ordinary polynomial.

$$z^2 H(z) = z^2 - 1.5z + 0.5 = (z-1)(z-0.5)$$
- 2 Zeros are at $z=1$ and $z=0.5$.
- 3 The z^{-2} form introduces two poles at the origin.
- 4 The zero at $z=1$ lies on the unit circle, so DC is nulled.



Key idea. A zero on the unit circle corresponds directly to a frequency that is completely removed.

Nulling filters

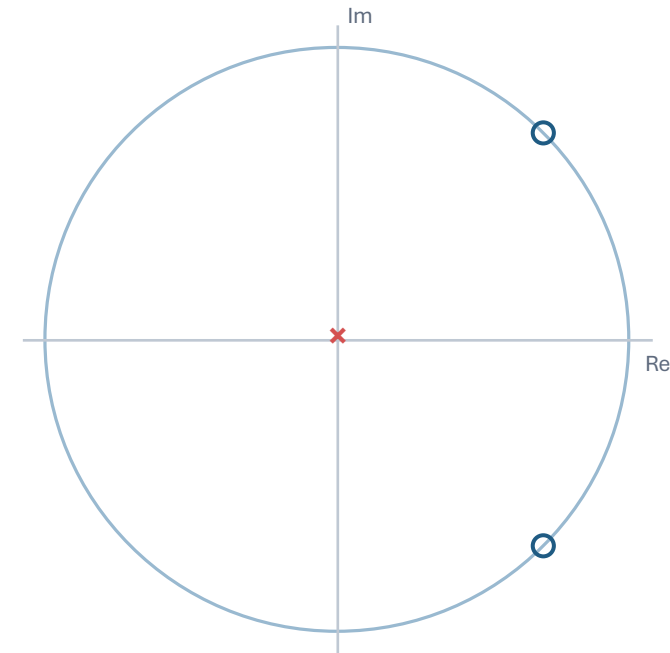
Place zeros on the unit circle at the unwanted frequency

$$H(z) = (1 - e^{j\hat{\omega}_0}z^{-1})(1 - e^{-j\hat{\omega}_0}z^{-1})$$

$$H(z) = 1 - 2\cos(\hat{\omega}_0)z^{-1} + z^{-2}$$

Why two zeros?

- A real cosine contains positive and negative frequency exponentials.
- To remove the real sinusoid, both conjugate components must be nulled.
- Real coefficients require conjugate-symmetric zeros.



Key idea. A sinusoid is suppressed when the unit-circle point at its normalized frequency coincides with a zero.

Example 4: design a sinusoid-nulling FIR filter

Suppress $x[n]=\cos(\pi n/4)$

1 The cosine contains frequencies $+\pi/4$ and $-\pi/4$.

2 Place zeros at $z=e^{j\pi/4}$ and $z=e^{-j\pi/4}$.

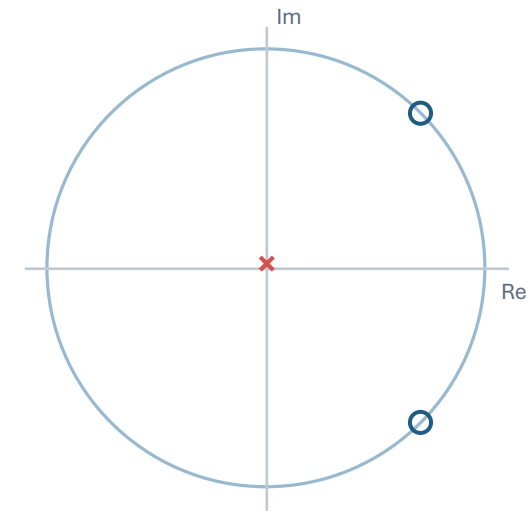
3 Use the conjugate-zero product.

$$H(z)=(1-e^{j\pi/4}z^{-1})(1-e^{-j\pi/4}z^{-1})$$

4 Expand using $2\cos(\pi/4)=\sqrt{2}$.

$$H(z)=1-\sqrt{2}z^{-1}+z^{-2}$$

$$y[n]=x[n]-\sqrt{2}x[n-1]+x[n-2]$$



Running-sum filters and roots of unity

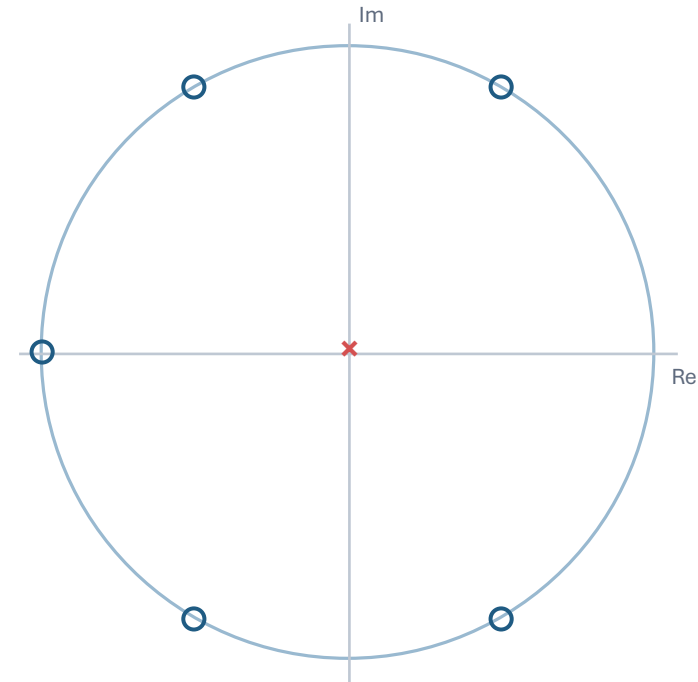
A simple FIR with regularly spaced unit-circle zeros

$$y[n] = \sum_{k=0}^{L-1} x[n-k]$$

$$H(z) = 1 + z^{-1} + \dots + z^{-(L-1)}$$

$$H(z) = (1 - z^{-L}) / (1 - z^{-1})$$

- Zeros are the L roots of unity except $z=1$, which cancels.
- The missing zero at $z=1$ leaves strong DC response.
- The remaining zeros produce exact nulls at normalized frequencies $2\pi k/L$.



Key idea. Running sums behave like low-pass filters with regularly spaced frequency nulls.

From running-sum to band-pass filters

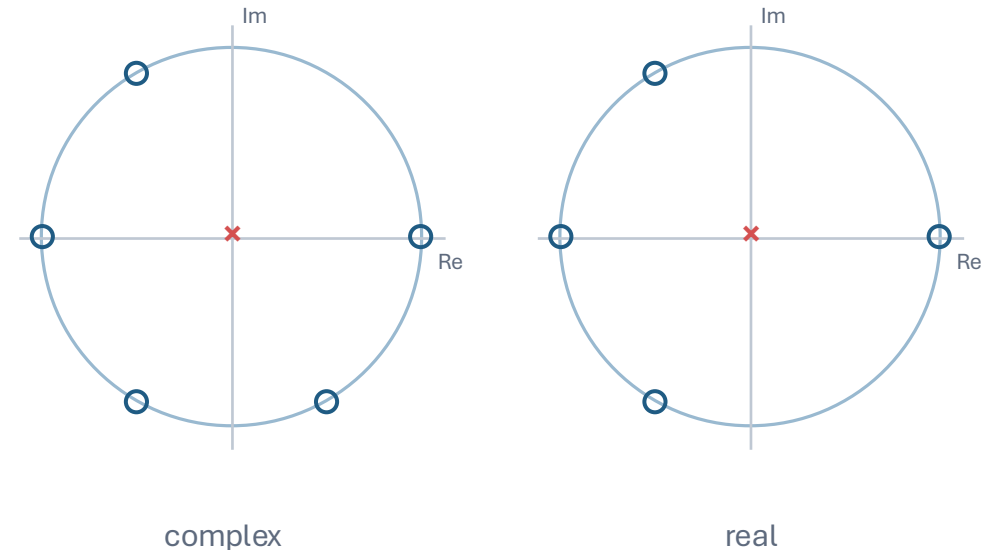
Omit a different root to move the passband

Complex band-pass:

- A running-sum-like zero pattern can be shifted.
- If one root is omitted at $+\hat{\omega}_0$, the response is largest near $+\hat{\omega}_0$.
- The coefficients are generally complex unless the zero pattern is conjugate symmetric.

Real band-pass:

- A real filter needs a matching passband at $-\hat{\omega}_0$.
- Conjugate symmetry gives real coefficients and equal magnitude lobes at $\pm\hat{\omega}_0$.



Key idea. Zero patterns on the unit circle can be shaped to pass selected frequency regions.

Linear-phase FIR systems

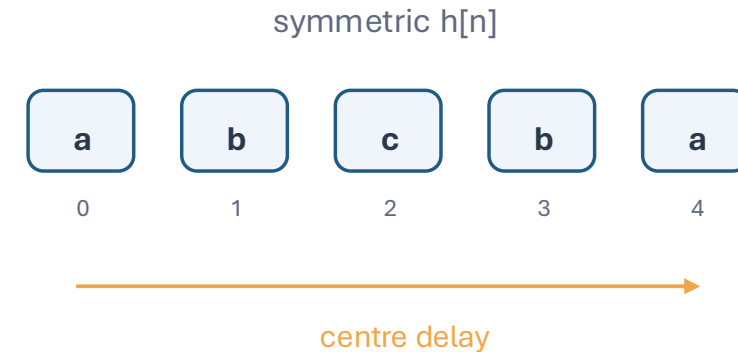
Symmetric coefficients produce a pure delay plus a real amplitude factor

$$h[n] = h[M-n]$$

$$H(e^{j\omega}) = e^{-j\omega M/2} A(\omega)$$

Interpretation:

- $A(\omega)$ is real, so it changes magnitude and may add π jumps.
- The factor $e^{-j\omega M/2}$ is a delay of $M/2$ samples.
- Waveform shape is preserved apart from filtering and delay.



Key idea. Linear phase is one of the main reasons FIR filters are widely used in audio and signal processing.

Notation and takeaways

Minimum language for z-transform analysis

$x[n]$	discrete-time sequence	$H(e^{j\hat{\omega}})$	frequency response on unit circle
$X(z)$	z-transform	zero	point where $H(z)=0$
$H(z)$	system function / transfer function	pole	point where $H(z)$ is undefined
z^{-1}	one-sample delay	unit circle	$z=e^{j\hat{\omega}}$

Three core ideas

- Time-domain delays become powers of z^{-1} .
- Convolution becomes multiplication of z-domain polynomials.
- Zeros and poles explain the frequency response and filter behaviour.